

Mission 2

La commande [dir flash:] permet d'afficher toutes les sauvegardes de configuration.

```
Router#dir flash:
Directory of flash0:/

 1  -rw-   103727964  Nov 22 2014 12:28:46 +00:00  c2900-universalk9-mz.SPA.154-3.M.bin
 2  -rw-     3064    Nov 22 2014 12:42:06 +00:00  cpconfig-29xx.cfg
 3  -rw-     2747    Apr 7 2016 14:53:46 +00:00  alexisz
 4  drw-       0    Nov 22 2014 12:42:46 +00:00  ccpepx
245 -rw-     2464    Nov 22 2014 12:44:38 +00:00  home.shtml
246 -rw-     4533    Apr 8 2016 08:44:50 +00:00  alexis
247 -rw-     1682   Dec 13 2016 09:08:38 +00:00  vpn
248 -rw-     1557    Feb 9 2017 10:23:16 +00:00  JordanClement
249 -rw-     2671   Oct 13 2017 08:34:14 +00:00  velon-gsb
250 -rw-     2806   Nov 24 2017 10:20:28 +00:00  velon-gsb-routeur
251 -rw-     2730   Jun 11 2019 06:57:20 +00:00  jbo-ppe-config
252 -rw-     4847   Nov 21 2019 16:05:52 +00:00  Conf-HPK-Thibaud_Mathys
253 -rw-     1656   Sep 25 2020 08:55:28 +00:00  laurentin-config
254 -rw-     3716   Nov 23 2020 15:59:04 +00:00  wifisepe
255 -rw-     3032   Mar 11 2021 15:57:46 +00:00  PENNE-SEGGARA
256 -rw-     5253   Apr 2 2021 07:40:08 +00:00  PENNE-SEGARRA
257 -rw-     1152   Sep 16 2021 13:05:36 +00:00  no
258 -rw-     4809   Nov 18 2021 14:49:52 +00:00  valentin-config
259 -rw-     2094   Mar 25 2022 09:48:14 +00:00  radius-config
260 -rw-     2918   Oct 17 2023 07:57:30 +00:00  Conf-HPK-LVA
261 -rw-     4692   Mar 5 2024 07:32:42 +00:00  Conf-HPK-LVA_YME
262 -rw-     3414   Nov 14 2023 09:15:34 +00:00  MaConfig2
263 -rw-     1674   Nov 23 2023 12:17:24 +00:00  TP-wifi
264 -rw-     5949   Mar 18 2025 15:27:56 +00:00  johan
265 -rw-     3353   Dec 3 2024 15:01:12 +00:00  config_switch
266 -rw-     2323   Oct 9 2025 14:15:38 +00:00  dalessandro-config-hpk
267 -rw-     3411   Nov 6 2025 15:17:18 +00:00  YAK-HPK-ConfRoute
```

La commande copy nom_de_la_sauvegarde running-config permet de restaurer la configuration en cours.

Installation de pfSense

Configuration des adresses IP de pfSense.

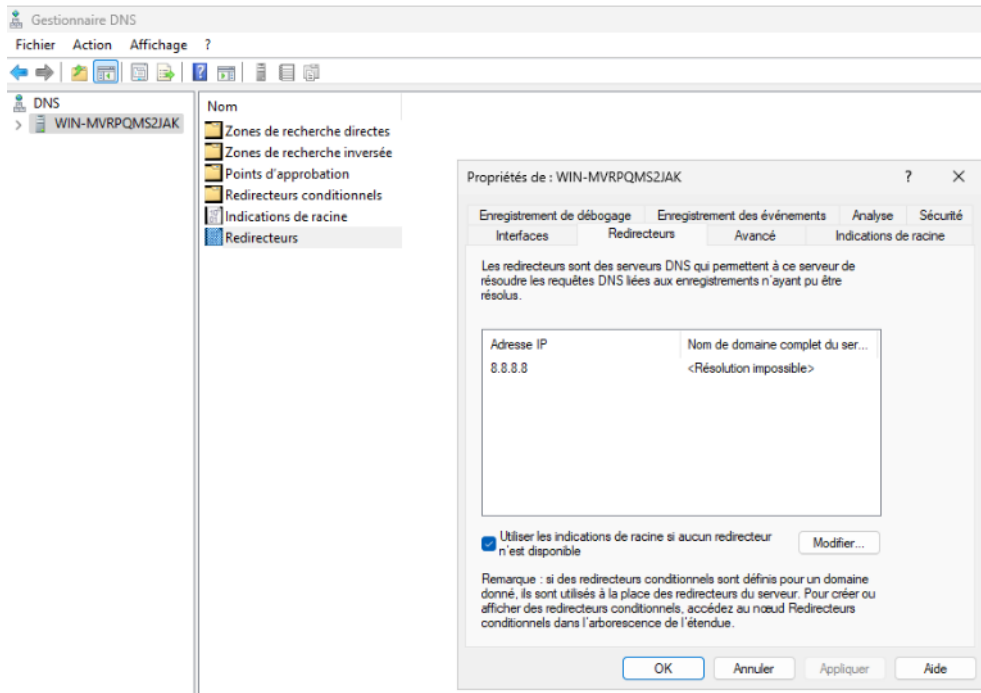
Création des access lists

Création de la route-map

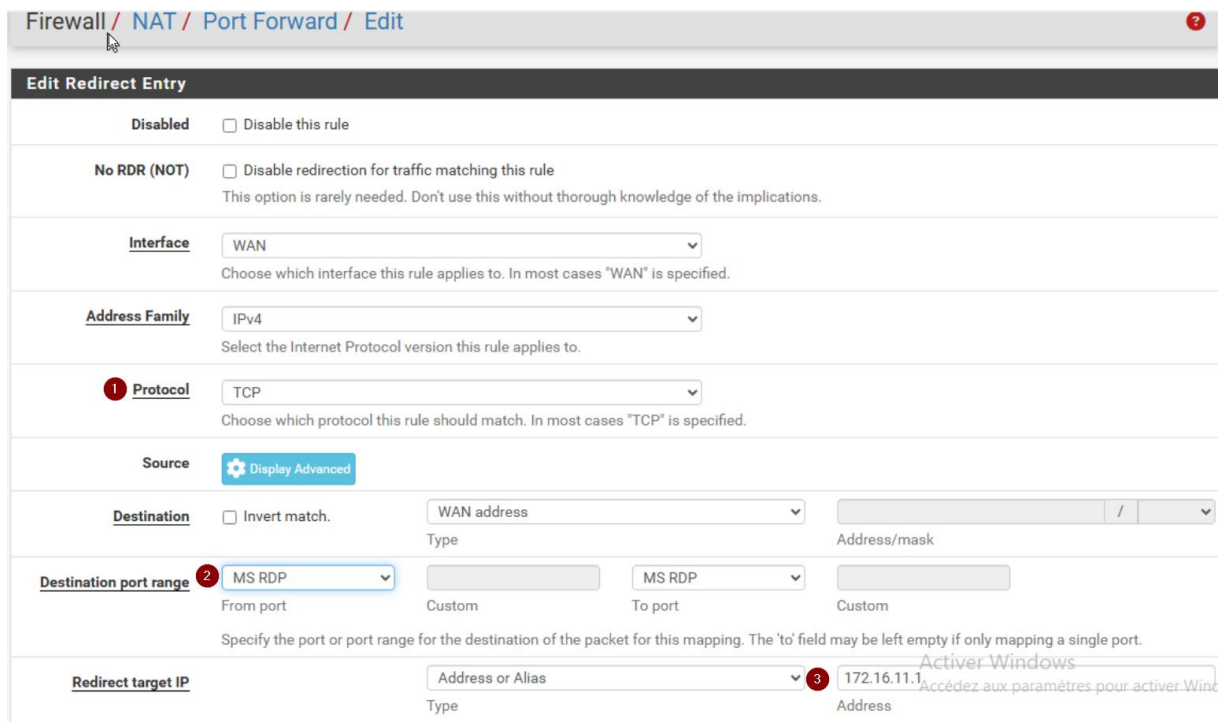
```
Router(config)#int g0/0.16
Router(config-subif)#route-map AKHOURFI permit
Router(config-route-map)#match ip address 95
Router(config-route-map)#set ip next-hop 172.18.11.1
Router(config-route-map)#ex
Router(config)#int g0/0.17
Router(config-subif)#route-map AKHOURFI permit
Router(config-route-map)#match ip address 95
Router(config-route-map)#set ip next-hop 172.18.11.1
Router(config-route-map)#ex
Router(config)#int g0/0.401
Router(config-subif)#route-map AKHOURFI permit
Router(config-route-map)#match ip address 95
Router(config-route-map)#set ip next-hop 172.18.11.1
Router(config-route-map)#ex
```

Application de la commande ip policy route-map AKHOURFI sur chaque interface.

Mise en place du redirecteur DNS



Ouverture du port 3389 de l'adresse IP SIO vers l'Active Directory.



Source			
Destination		☐ Invert match.	WAN address Type
Destination port range		MS RDP From port	MS RDP To port
Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.			
Redirect target IP		Address or Alias Type	172.16.11.1 Address
Enter the internal IP address of the server on which to map the ports. e.g.: 192.168.1.12 for IPv4 In case of IPv6 addresses, it must be from the same "scope", i.e. it is not possible to redirect from link-local addresses scope (fe80:*) to local scope (::1)			
Redirect target port		MS RDP Port	Custom
Specify the port on the machine with the IP address entered above. In case of a port range, specify the beginning port of the range (the end port will be calculated automatically). This is usually identical to the "From port" above.			
Description			
A description may be entered here for administrative reference (not parsed).			

Rattachement de pfSense à l'Active Directory

Création du certificat en conservant les paramètres par défaut.

System / Certificates / Authorities / Edit

Authorities

Certificates

Revocation

Create / Edit CA

Descriptive name

YAK-HPK-CSSL

The name of this entry as displayed in the GUI for reference.
This name can contain spaces but it cannot contain any of the following characters: ?, >, <, &, /, \, " , ' ,

Method

Create an internal Certificate Authority

Trust Store

☐ Add this Certificate Authority to the Operating System Trust Store

When enabled, the contents of the CA will be added to the trust store so that they will be trusted by the operating system.

Randomize Serial

☐ Use random serial numbers when signing certificates

When enabled, if this CA is capable of signing certificates then serial numbers for certificates signed by this CA will be automatically randomized and checked for uniqueness instead of using the sequential value from Next Certificate Serial.

Internal Certificate Authority

Key type

RSA

2048

The length to use when generating a new RSA key, in bits.
The Key Length should not be lower than 2048 or some platforms may consider the certificate invalid.

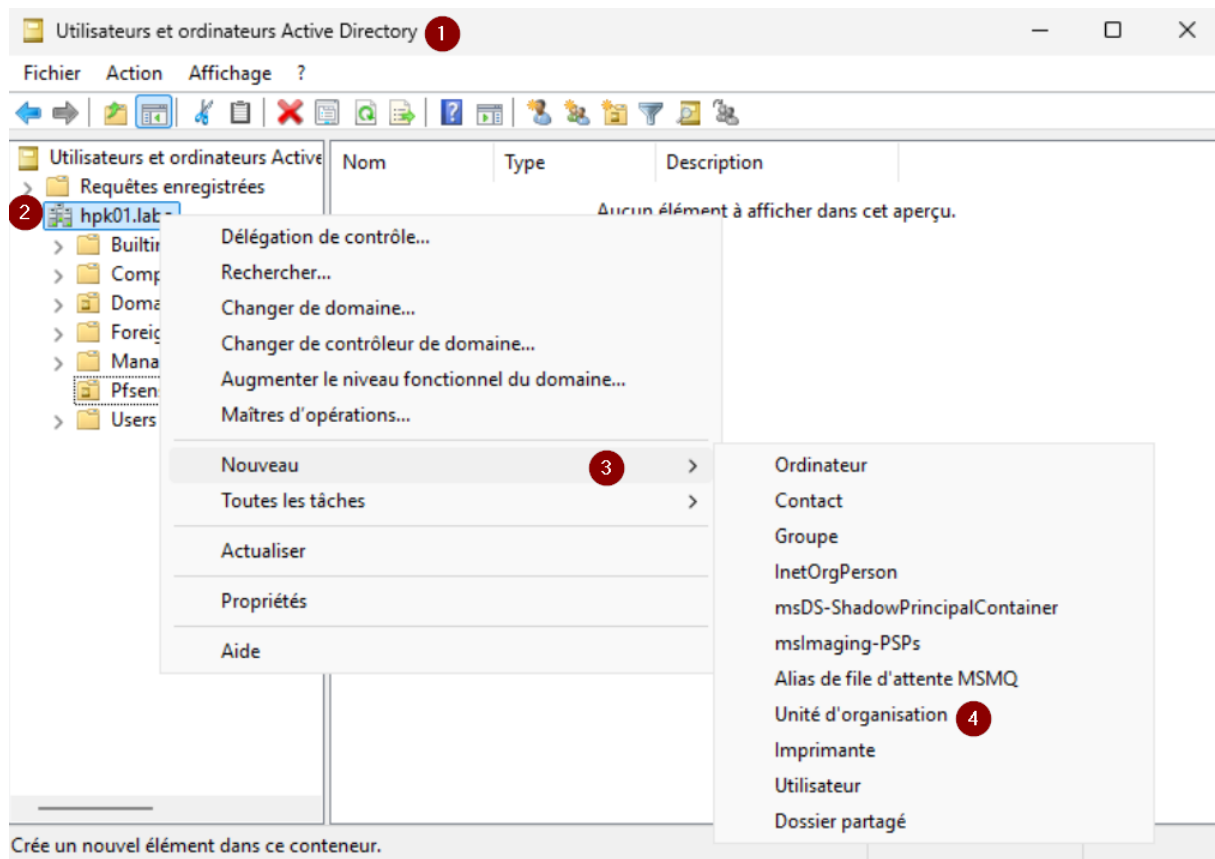
Digest Algorithm

sha256

The digest method used when the CA is signed.
The best practice is to use SHA256 or higher. Some services and platforms, such as the GUI web server and OpenVPN, consider weaker digest algorithms invalid.

Activar Windows

Activar Windows



Ajout du rôle « Services de certificats Active Directory ».

Création d'un utilisateur dans l'UO.

Création de l'utilisateur et du groupe.

